20/12/24 MIS

USN | | | | | | | | | | |

[22AS561]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

COMPOSITE MATERIALS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	Why combine materials to make a Composite?	L1	CO1	**02**
	b.	What is Kevlar fiber?	L1	CO2	**02**
	c.	Explain different forms of glass fibers.	L2	CO2	**02**
	d.	What is ILSS? Explain the Importance of ILSS.	L1	CO1	**02**
	e.	What is fatigue loading? What is the stress ratio?	L1	CO1	**02**
		PART-B		**6 X 5=30**	
2.	**a.**	Explain the tensile and compression tests for composite specimens with neat sketches.	L2	CO1	**05**
		OR			
	b.	Discuss the various types of Composites with the help of tables and sketches.	L2	CO1	**05**
3.	**a.**	Explain how the Kevlar fiber composites are better for ballistic impact, pressure vessel, damping, cut-proof and fireproof structures.	L2	CO3	**05**
		OR			
	b.	Explain the manufacturing process of Ceramic matrix composites by hot pressing of powders with neat sketches.	L2	CO4	**05**
4.	**a.**	Explain the Pultrusion process for composite fabrication. List the applications and advantages of pultrusion	L2	CO3	**05**
		OR			
	b.	Explain how backscattering-type ultrasonic inspection can be used to find fiber damages and waviness. Explain its limitations and applications.	L2	CO6	**05**
5.	**a.**	List the advantages, limitations and applications of ceramic matrix composites.	L1	CO4	**05**
		OR			
	b.	Explain the defects in a laminated honeycomb structure with a neat sketch. Explain the possible non-destructive methods of inspection.	L2	CO6	**05**

Deemed-to-be University Learning Resource Centre Jain Global Campus Ramanagar-562

6.	a.	Explain the diffusion bonding method for fabrication of metal matrix composites.	L2	CO3	05
		OR			
	b.	Explain Thermal barrier coating with air plasma process.	L2	CO5	05
7.	a.	Explain the eddy current test for surface cracks with a neat sketch. List the applications and advantages of eddy current technology.	L2	CO6	05
		OR			
	b.	What is BVID? Explain impact damage cone in impacted laminated composites. Explain the stages of failure in fatigue loading for composites.	L1	CO1	05
		PART-C		**6 X 10=60**	
8.	a.	What is the Shear strength of a material? Explain two-rail and three-rail shear tests for composite plates with neat sketches.	L2	CO 1	10
		OR			
	b.	Identify the important structural parts and material selection of a light aircraft.	L2	CO 1	10
9.	a.	Explain the applications and properties of carbon fiber. Explain the manufacture of carbon fiber from PAN (Poly Acrylo Nitrile) fibers with a neat sketch.	L2	CO 2	10
		OR			
	b.	What are Thermal Barrier Coatings? Explain the different types of materials used and the conduction of heat in TBC layers with a neat sketch.	L2	CO5	10
10.	a.	Explain Continuous fiber ceramic matrix composites by sol-gel process. List the advantages and limitations.	L2	CO4	10
		OR			
	b.	Explain the fabrication of carbon-carbon composites by the Pyrolysis process. Explain how space shuttles are protected from extremely high temperatures of re-entry.	L2	CO3	10
11.	a.	Explain the strength and stiffness properties of glass fiber composites with neat sketches. Also discuss how different forms of fiber will affect the strength and stiffness of structures.	L2	CO3	10
		OR			
	b.	List the important structural parts of a large aircraft with the help of figures.	L1	CO1	10

12.	**a.**	What is radiography? What are the basic steps in radiography? List the advantages, limitations and applications.	L1	CO6	**10**
		OR			
	b.	What is carbon-carbon composite? Explain its properties and applications. Explain the process of making carbon-carbon composites by CVI (chemical vapor infiltration) process.	L2	CO3	**10**
13.	**a.**	Explain how a laminated composite structure is fabricated with neat sketch. List the different types of defects in laminated structures and possible methods of NDT for inspection.	L2	CO6	**10**
		OR			
	b.	Explain the eddy current test for surface cracks with a neat sketch. List the applications and advantages of eddy current technology.	L2	CO6	**10**